

Generalized diffusion and stylized facts for asset prices

Peter Richmond^a and Stefan Hutzler^b

^aBusiness School, Trinity College,
Dublin 2, Ireland
peter_richmond@btinternet.com

^bSchool of Physics, Trinity College
Dublin 2, Ireland
stefan.hutzler@tcd.ie

Abstract

We show how generalized diffusion coefficients determine both the Fokker Plank equations and equations of motion for statistical systems. In passing we note the approach yields a generalization of Ito's stochastic calculus that is not restricted to second order differentials. For financial systems, the approach can be used to compute not only the probability density distribution function of log price returns but also the autocorrelation functions for linear and quadratic log price returns.

Using both high frequency and daily data for a number of different assets we compute the generalized diffusion coefficients and show how the theoretical framework then leads to the well known stylized facts including fat tails, exponential decay for linear return correlations and slow power law decay for square return correlations.

Keywords

generalized diffusion, asset prices, stylized facts